GSE 518
Prof. Carlos A. Flores
Cal Poly State University

Suggested Solutions to Problem Set 9

1.
(a) For v^2 we have

$$\begin{aligned} E\left(v^2\right) &= E\left(\frac{1}{n}\sum_{i=1}^{n} X_i^2\right) \\ &= \frac{1}{n}\sum_{i=1}^{n} E\left(X_i^2\right) \\ &= \frac{1}{n}\sum_{i=1}^{n} \sigma_0^2 = \frac{n\sigma_0^2}{n} = \sigma_0^2 \end{aligned}$$

Therefore, v^2 is unbiased. As for s^2, we showed in class that $E\left(s^2\right) = \sigma_0^2$, so s^2 is also unbiased.
(b) Since both estimators are unbiased, we have that $MSE\left(v^2\right) = var\left(v^2\right)$ and $MSE\left(s^2\right) = Var\left(s^2\right)$. We now calculate the variance of each of these estimators.
From question 3 in problem set 7 we know $Var\left(s^2\right) = \frac{2\sigma_0^4}{n-1}$. We now derive the variance of v^2. Remember that $X_1, X_2, \ldots, X_n$ are iid with distribution $N\left(0, \sigma_0^2\right)$. Therefore, we know that $\frac{X_i}{\sigma_0} \sim N\left(0,1\right)$. Moreover, since the X_i are independent we have that

$$\sum_{i=1}^{n}\left(\frac{X_i}{\sigma_0}\right)^2 \sim \chi^2_{(n)}$$

For this result, remember that the square of a standard-normally distributed random variable follows a Chi-squared distribution with one degree of freedom, at that the sum of independent Chi-squared random variables also follows a Chi-squared distribution with the degrees of freedom equal to the sum of the degrees of freedom (e.g., see Lemma 5.3.2 in CB and question 5 in problem set 6). Therefore, we have that

$$\frac{nv^2}{\sigma_0^2} = \sum_{i=1}^{n}\left(\frac{X_i}{\sigma_0}\right)^2 \sim \chi^2_{(n)}$$

We know that the variance of a $\chi^2_{(r)}$ equals $2r$. Therefore, following steps similar to those in question 3 in problem set 7, we have that $Var\left(\left(n\right)v^2/\sigma_0^2\right) = 2n$. Hence we have

$$\begin{aligned} 2n &= Var\left(\frac{nv^2}{\sigma_0^2}\right) \\ 2n &= \frac{\left(n\right)^2}{\sigma_0^4} Var\left(v^2\right) \end{aligned}$$

which implies

$$Var\left(v^2\right) = \frac{2\sigma_0^4}{n}$$

Therefore, we have that $Var(s^2) = \frac{2\sigma_0^4}{n-1} > \frac{2\sigma_0^4}{n} = Var(v^2)$. Therefore, v^2 has a lower MSE.

2. From Corollary 7.3.10 or the Cramèr-Rao Inequality theorem for the iid case we saw in class, we have that the Cramèr-Rao Lower Bound for any unbiased estimator $\widehat{\theta}$ is given by

$$Var(\widehat{\theta}) \geq \frac{1}{nI(\theta_0)}$$

where $I(\theta_0)$ is the Fisher's Information matrix and equals

$$I(\theta_0) = Var(S(X,\theta_0)) = E\left[S^2(X,\theta_0)\right] = -E\left[\frac{\partial^2 \ln f(x,\theta_0)}{\partial\theta^2}\right] \tag{1}$$

(a) (i) From problem set 8 question 2(a) we know $\ln f(x,\theta_0)$ is given by

$$\ln f(x|\theta) = x\ln(\theta) - \theta - \ln(x!)$$

Hence the score function is given by

$$S(X,\theta) = \frac{\partial \ln f(x,\theta)}{\partial\theta} = \frac{1}{\theta}X_i - 1$$

We can use any of the formulas in (1) to calculate $I(\theta_0)$. Here is important to remember that for $Poisson(\theta_0)$ we have $E(X) = \theta_0$ and $Var(X) = \theta_0$. Hence we have

$$\begin{aligned} I(\theta_0) &= Var(S(X,\theta_0)) \\ &= Var\left(\frac{1}{\theta}X_i - 1\right) \\ &= \frac{1}{\theta_0^2}Var(X_i) \\ &= \frac{\theta_0}{\theta_0^2} = \frac{1}{\theta_0} \end{aligned}$$

Therefore, in this case the CRLB is

$$CRLB = \frac{1}{nI(\theta_0)} = \frac{\theta_0}{n}$$

(ii) From problem set 8 question 2(a), the MLE estimator for θ_0 is $\widehat{\theta}_{MLE} = \overline{X}$. By the properties of $\overline{X}$ and given $X \sim Poisson(\theta_0)$, we have $E\left(\overline{X}\right) = E(X_i) = \theta_0$ and $Var\left(\overline{X}\right) = Var(X_i)/n = \theta_0/n$. Since $\widehat{\theta}_{MLE}$ is unbiased and its variance equals the $CRLB$ (i.e., $Var(\widehat{\theta}_{MLE}) = CRLB = \frac{\theta_0}{n}$), we have that $\widehat{\theta}_{MLE} = \overline{X}$ is the UMVUE for θ_0.

(b) (i) From problem set 8 question 2(b) we know $\ln f(x,\theta_0)$ is given by

$$\ln f(x|\theta) = -\ln(\theta) - \frac{1}{\theta}x$$

Hence the score function is given by

$$S(X,\theta)=\frac{\partial \ln f(x,\theta)}{\partial \theta}=-\frac{1}{\theta}+\frac{1}{\theta^2}X_i$$

We can use any of the formulas in (1) to calculate $I(\theta_0)$. Here is important to remember that for $Exponential(\theta_0)$ we have $E(X)=\theta_0$ and $Var(X)=\theta_0^2$. Hence we have

$$\begin{aligned} I(\theta_0) &= Var(S(X,\theta_0)) \\ &= Var\left(-\frac{1}{\theta_0}+\frac{1}{\theta_0^2}X_i\right) \\ &= \frac{1}{\theta_0^4}Var(X_i) \\ &= \frac{\theta_0^2}{\theta_0^4}=\frac{1}{\theta_0^2} \end{aligned}$$

Therefore, in this case the CRLB is

$$CRLB=\frac{1}{nI(\theta_0)}=\frac{\theta_0^2}{n}$$

(ii) From problem set 8 question 2(b), the MLE estimator for θ_0 is $\widehat{\theta}_{MLE}=\overline{X}$. By the properties of $\overline{X}$ and given $X\sim Exponential(\theta_0)$, we have $E\left(\overline{X}\right)=E(X_i)=\theta_0$ and $Var\left(\overline{X}\right)=Var(X_i)/n=\theta_0^2/n$. Since $\widehat{\theta}_{MLE}$ is unbiased and its variance equals the $CRLB$ (i.e., $Var(\widehat{\theta}_{MLE})=CRLB=\frac{\theta_0^2}{n}$), we have that $\widehat{\theta}_{MLE}=\overline{X}$ is the UMVUE for θ_0.

3.

(a) Using the density we can calculate $E(X)=\int_0^{\theta_0}x\frac{2x}{\theta_0^2}dx=\frac{2}{3}\theta_0$. Hence we have

$$E(\widehat{\theta}_1)=E\left(\frac{3\overline{X}}{2}\right)=\frac{3E\left[\overline{X}\right]}{2}=\frac{3}{2}\left(\frac{2}{3}\theta_0\right)=\theta_0$$

So $\widehat{\theta}_1$ is an unbiased estimator for θ_0.

(b) In this case we have that $\ln f(x,\theta_0)$ is given by

$$\ln f(x|\theta)=\ln(2x)-2\ln(\theta)$$

Hence the score function is given by

$$S(X,\theta)=\frac{\partial \ln f(x,\theta)}{\partial \theta}=-\frac{2}{\theta} \tag{2}$$

In this case the assumptions of the CRLB does not hold, and the same is true about the assumptions regarding the Score Identity and the Information Matrix Identity. Hence, in

this case each formula in (1) gives a different result. Here we use the original formula for the CRLB (see Corollary 7.3.10), which states that the CRLB equals

$$CRLB = \frac{1}{nE\left[S^2\left(X, \theta_0\right)\right]}$$

Using (2) to calculate $E\left[S^2\left(X, \theta_0\right)\right]$ we find

$$\begin{aligned} E\left[S^2\left(X, \theta_0\right)\right] &= E\left[\left(-\frac{2}{\theta_0}\right)^2\right] \\ &= E\left[\frac{4}{\theta_0^2}\right] \\ &= \frac{4}{\theta_0^2} \end{aligned}$$

Therefore, in this case the CRLB is

$$CRLB = \frac{1}{nE\left[S^2\left(X, \theta_0\right)\right]} = \frac{\theta_0^2}{4n}$$

(c) Using the density we can calculate $E\left(X^2\right) = \int_0^{\theta_0} x^2 \frac{2x}{\theta_0^2} dx = \frac{1}{2}\theta_0^2$, and therefore, $Var\left(X\right) = E\left(X^2\right) - \left[E\left(X\right)\right]^2 = \frac{1}{18}\theta_0^2$. Hence we have

$$\begin{aligned} Var(\widehat{\theta}_1) &= Var\left(\frac{3\overline{X}}{2}\right) \\ &= \frac{9Var\left(\overline{X}\right)}{4} \\ &= \frac{9\theta_0^2}{4\left(18\right)n} \\ &= \frac{\theta_0^2}{8n} \end{aligned}$$

where in the third line we used the fact that $Var\left(\overline{X}\right) = Var\left(X_i\right)/n = \frac{1}{n18}\theta_0^2$.
Therefore, we have that $Var(\widehat{\theta}_1) = \frac{\theta_0^2}{8n} < CRLB = \frac{\theta_0^2}{4n}$. As stated in the problem set, this happens because the assumptions of the CRLB are not satisfied in this case.